

Technology Stack

Date 29/09/2026

Team ID SWTID-2026-6174

Project Name FitBuddy – AI Fitness Plan Generator using Gemini Models

Maximum Marks 2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Jinja2	Used to create the user input form, personalized result page, feedback form, and admin view. Jinja2 enables dynamic rendering of FitBuddy data and AI-generated results.
2	Backend / Server-Side	Python + FastAPI	FastAPI handles routing, form processing, validation, Gemini API calls, and communication between the frontend, AI layer, and database.
3	Database / Data Storage	SQLite + SQLAlchemy	SQLite stores user information and generated plans. SQLAlchemy ORM provides structured database access and persistence for original and updated plans.

Technology Stack Details (continued)

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	Cloud / Hosting / Deployment	Uvicorn + Local Deployment	Uvicorn runs the FastAPI application as an ASGI server for local deployment and testing. The project documentation describes local execution and notes future cloud deployment possibilities.
5	Version Control & CI/CD	Git	Git is used for version control and project collaboration. It helps track source-code changes and maintain different versions of the FitBuddy application.

Project Environment & Supporting Tools

Python and pip are used for the development environment. The project uses a virtual environment and installs FastAPI, Uvicorn, Jinja2, SQLAlchemy, python-multipart, and the Google Generative AI SDK. The application is tested locally through the FastAPI/Uvicorn server.

Technology Stack Details (continued)

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools	Google Gemini API (Gemini 1.5 Pro & Gemini Flash)	Gemini 1.5 Pro generates structured 7-day workout plans and updates them using user feedback. Gemini Flash generates fast nutrition or recovery tips. The API integration connects these AI capabilities with FastAPI.

Technology Stack Summary

Frontend: HTML, CSS and Jinja2 **Backend:** Python and FastAPI **Database:** SQLite with SQLAlchemy

Deployment: Uvicorn local server **Version Control:** Git **AI API:** Google Gemini 1.5 Pro and Gemini Flash